# The Field Equations of the Screened Khronon Action, and a Verdict for Every Physics Class

Working note (PAPER70), v1.5, 2026-11-08 (v1.0 deposited as DOI 10.5281/zenodo.24040313; v1.1, which added the kernel erratum of Section 6, deposited as DOI 10.5281/zenodo.24044606; v1.2 deposited as DOI 10.5281/zenodo.24047420; v1.3, which carried the owner's acceptance of the corrected-kernel labels, deposited as DOI 10.5281/zenodo.24047508; v1.4, which added the one-page final theory to the supplement with the paper unchanged, deposited as DOI 10.5281/zenodo.24047634; this version adds the anatomy of the CMB requirement, Section 7).
AI-assisted research programme; not peer reviewed. Companions: PAPER67 (DOI 10.5281/zenodo.23968171, the law and its failure record), PAPER68 (DOI 10.5281/zenodo.23990944, the action) and PAPER69 (DOI 10.5281/zenodo.24025170, the closure record). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

**Abstract**

Gravity carries one extra scale, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, $\kappa = \frac{1}{2}$ fitted (proved not derivable), realised by the v3s screened khronon action. It passes Cassini, PPN, GW speed, galaxy lensing and SPARC, and fails AGC 114905, Malin 1 and the canonical FG1 tilt. It also under-fills X-COP clusters, groups, bright M31 dSphs and Coma UDGs by amounts that vary about 100× between classes, after the IMF accounts for the ellipticals and most of KiDS. Separately, it fails the CMB-lensing shape, which needs cold extra gravity at $z$ 3400. No cold fluid, fixed-temperature sea, thermal light fermion or single measured key tried so far supplies either. The next lane must find one covariant quantity that sets that amplitude (not just a switch) in today's hot systems and is non-zero in the early universe, or show the two failures need separate causes. *(This is the checked seed paragraph of 2026-11-06, verbatim.)*

Since the seed, a joint systematics budget per class also explains the bright M31 dSphs, and a symmetric admissibility cut leaves the Coma UDGs NOT A TEST, so the measured boundary of the law is the X-COP clusters, the groups (fragile) and the CMB-lensing shape; and a shock-entropy sector written into the action fails by the no-tuning rule, so every physics class on the record carries a label. This note writes down the action in the Legendre form every lane solved (correcting the form printed in PAPER68), its complete covariant field equations checked against direct variation, their 3+1 form and a convergent static solver. It then lists every gate with its frozen label, states the residue as measured numbers, and collects the three frozen tests that can kill the law. Four physical classes of extra ingredient were built and graded on both footings: a cold fluid, a warm fixed-temperature sea, a hot thermal fermion, and a shock-entropy sector written into the action. Each meets its own wall, and no candidate passes every gate. The most probable reading of the record is the deposited, scoped law plus separate per-class systematics, with the cluster, group and ultra-diffuse rows and the CMB wall left unexplained.

## 1 Conventions

Units $c = 1$ and $K = 4\pi G$ unless stated; restoring $c$: $K \to 4\pi G/c^2$, the prefactor of $R$ is $c^4/16\pi G$, and $a_0 \to a_0/c^2$ inside $W$. Two footings are carried and never pooled: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. $\kappa$ is fitted; a Lean proof shows that it is not derivable from the programme's premises. A gate is closed when it carries a label fixed by criteria committed before the first graded number; every lane runs a mutation twin that must fail.

## 2 The action, in Legendre form

PAPER68 prints $S = \frac{1}{4K}\int\sqrt{-g}\,[R - 2\mathcal{J}(Y) + 2\mathcal{K}(Q)] - \frac{\gamma_L}{2}\int\sqrt{-g}\,[\max(D\cdot P, 0)]^2 + S_m$. Read literally, $P$ then appears only in the stiffness, its own equation forces $D\cdot P$ to be constant, and $P$ decouples from gravity. The form every lane has actually solved puts $\mathcal{J}$ in Legendre form (QX17):

$$\frac{\mathcal{L}}{\sqrt{-g}} = \frac{R - 2\Lambda + 2\mu_K^2(Q-1)^2}{4K} + P^m A_m - W(K|P|) - \gamma_L F(\Pi) + \lambda\, n^m P_m, \qquad W(p) = \frac{a_0^2}{K}\Big[-\frac{p}{a_0} - \ln\Big(1 - \frac{p}{a_0}\Big)\Big], \quad (1)$$

with $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$, $\Pi = D_m P^m$ and $\gamma_L = KL_\star^2/2$, plus minimally coupled matter $S_m$. Maximising over $P$ returns the printed $\mathcal{J}$ exactly: $\max_P[P\cdot A - W(K|P|)] = (a_0^2/K)[u - \ln(1+u)]$ with $u = |A|/a_0$, attained at $K|P| = a_0 u/(1+u)$ (sympy, residual 0). So at $\gamma_L = 0$, Eq. (1) is the printed action; with $\gamma_L > 0$, $P$ is an

independent field and Eq. (1) is the theory. **This is an erratum on PAPER68's printed form only; no number changes.**

**Adopted stiffness (owner decision, 2026-11-07).** The owner adopted the $\Pi^6$ blend as the action's stiffness. In Eq. (1) the raw kink $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$ is replaced by AC21 iter_6's blend, $F(\Pi) = b^2 f(\Pi/b)$ with $b = \rho_{\rm DE}$, the $a_0$ law's own scale, so there is no new constant. Near 0, $f$ is a pure power, $f(q) = (122363904/44610575)\, q^6$ for $q \le \frac{1}{2}$ ($m = 6 > d + 2 = 5$). On the layer it is a $C^4$ convex blend, and $f(q) = q^2/2$ exactly for $q \ge 1$, so $F = \frac{1}{2}\Pi^2$ for $\Pi \ge b$. Eq. (1) with this $F$ is the operative action of this note; it supersedes PAPER68's printed raw form. AC2q is not adopted; if it is adopted later, it carries the same blend with $b = X$. The scope of v3s and every graded label are unchanged: the blend moves no failing gate (largest effect $\le 1.4 \times 10^{-4}$ dex). The field equations of the next section hold for any stiffness $F$ with $\Pi_+$ read as $F'(\Pi)$ throughout and the stress-tensor term written $\Pi F'(\Pi)\, h_{mn}$; for the raw $F$ this is $2F h_{mn}$, as QX17 certified. QX17D certified the general form against a direct variation, for a generic polynomial $F$ and for the blend's pure-power branch (residuals $\le 7.1 \times 10^{-40}$); the shortcut that keeps $2F h_{mn}$ fails the metric equation by up to 14 %. Limits adopted with the blend: well-posedness is local in time; it is tested in spherical 3D only, and non-radial ($l \ge 1$) modes are untested; folds are unchanged; power 6 is not forced (the smallest integer with $m > d + 2$, admissible, not unique). The AC16 row remains the label of the raw switch.

**Definitions.** $\tau$ is the khronon; $Q = \sqrt{-g^{mn}\partial_m\tau\,\partial_n\tau}$ ($= 1/N$ in unitary gauge); $n_m = -\partial_m\tau/Q$; $h_{mn} = g_{mn} + n_m n_n$; $l = \ln Q$; $A_m = n^n\nabla_n n_m = -h_m{}^n\partial_n l$; $D_m = h_m{}^n\partial_n$; $\Pi = \nabla_m P^m - P^m A_m$; $\Pi_+ = \max(\Pi, 0)$; $\hat{P} = P/|P|$; $\mathcal{D}_m = \partial_m\Pi_+ - \Pi_+\partial_m l$.

# 3 The field equations (QX17, CERTIFIED)

$$\text{(E1)}\quad n^m P_m = 0, \tag{2}$$

$$\text{(E2)}\quad (1 + \gamma_L\Pi_+)A_m + \gamma_L D_m\Pi_+ = KW'(K|P|)\,\hat{P}_m, \qquad \lambda = (1 + \gamma_L\Pi_+)\, n\cdot\partial \ln N + \gamma_L\, n\cdot\partial\Pi_+, \tag{3}$$

$$\text{(E3)}\quad \nabla_m\big[n^m\rho_Q - \lambda P^m/Q\big] = 0, \qquad \rho_Q = \frac{\nabla_m[(1 + \gamma_L\Pi_+)P^m]}{Q} + \frac{\mu_K^2(Q-1)}{K}, \tag{4}$$

$$\text{(E4)}\quad G_{mn} + \Lambda g_{mn} = 2K\big(T^{\rm mat}_{mn} + T^Q_{mn} + T^W_{mn} + T^P_{mn}\big), \tag{5}$$

with

$$T^Q_{mn} = g_{mn}\frac{\mu_K^2(Q-1)^2}{2K} + \frac{\mu_K^2}{K}Q(Q-1)\, n_m n_n, \qquad T^W_{mn} = -g_{mn}W + KW'\,\frac{P_m P_n}{|P|},$$

$$T^P_{mn} = g_{mn}(P\cdot A) + 2P_{(m}\partial_{n)}l + (\nabla\cdot P)\, n_m n_n - 2\lambda\, n_{(m}P_{n)} + \gamma_L\big[-g_{mn}F + \Pi F'(\Pi)\, h_{mn} + h_{mn}P^r\mathcal{D}_r - 2P_{(m}\mathcal{D}_{n)}\big].$$

(E5) Matter is minimally coupled, $\nabla^m T^{\rm mat}_{mn} = 0$. The energy density seen by the foliation is $n^m n^n T_{mn} = (\mu_K^2/2K)(Q^2 - 1) + W + \Pi + \gamma_L F$. At $\gamma_L = 0$ the leaf part of (E2) inverts to $K|P| = a_0|E|/(a_0 + |E|)$, the Legendre map; $\Pi_+$ is $C^1$, so the equations hold in the weak sense across the switch surface.

**Checks.** Each equation is compared with a direct Euler–Lagrange variation of Eq. (1) on jet symbols (mpmath, 40 digits, three random configurations in 2+1 dimensions): P equation max rel $3.8 \times 10^{-41}$, khronon equation $1.8 \times 10^{-41}$, full stress tensor $1.2 \times 10^{-40}$; FRW reduction exact. The MUTATE twin (two flipped signs) fails at 0.17–0.78.

**AC2q** (candidate, not adopted; QX17A, CERTIFIED). The dark-energy scalar is the clock, there is no $\mathcal{K}(Q)$ term, and the switch acts on the khronon charge $\Sigma = \Pi + Q^2$: $\mathcal{L}/\sqrt{-g} = R/4K + X - V(\phi) + P\cdot A - W - \gamma\{F(\Pi + 2X) - (2X)^2/2\} + \lambda n\cdot P$, $X = Q^2/2$. Its P equation is (E2) with $\Pi_+ \to \Sigma_+$; its clock equation is $\nabla_m[n^m\rho_\phi - \lambda P^m/Q] = -V'(\phi)$ with $\rho_\phi = J_P/Q + Q[1 - 2\gamma(\Sigma_+ - Q^2)]$, which is $\Box\phi = V'$ at $P = \gamma = 0$; at $P = 0$ the stiffness stress vanishes identically, so the background is the canonical scalar's. The MUTATE twin fails at $0.23/0.39/7.3 \times 10^{-3}$.

# 4 3+1 form, reductions and solutions

**ADM** (QX17B). $P$ and $N$ carry no time derivatives, so the P equation and the lapse equation are constraints on each leaf. The P sector's momentum density vanishes on shell, so the momentum constraint is GR's; the tensor

principal part is GR's, so $c_T = c$. The Lean kernel file for these steps has 12 theorems, standard axioms only, and its MUTATE twin fails. The lane as a whole is not certified: its numerical ADM row failed as frozen on a sign convention in the checker's off-shell momentum density, and its `solve_bvp` spherical solver did not converge.

**Reductions.** FRW with $P = 0$: $\bar{Q} = 1 + I_0/(2\mu_K^2 a^3)$ and a dust-like $\rho \propto a^{-3}(1 + w_0/a^3)$. Static, weak, $\gamma_L = 0$: $\nabla \cdot [\mu(|\nabla\Phi|/a_0)\nabla\Phi] + \mu_K^2\Phi = 4\pi G\rho_b$, $\phi = \psi$. Spherical, $\mu_K = 0$: the law $g_N = g^2/(g + a_0)$ (Lean). Static, $\gamma_L > 0$: the Cassini hedgehog inside $L_\star$. AC2q FRW: the lapse–polarization mixing is $(1 - z)$ with $z = \gamma\dot{\phi}^2/\bar{N}^2$, so eliminating $P$ multiplies the induced lapse term by $(1 - z)^2$, not the legacy $(1 - 2z)^2$ (which is the exact factor of a different action). The UV speed is $c^2 = z/(1 - z + z^2)$, and $0 < c_g \le c_{\rm ph} \le 1$ holds for any $f \ge 0$, so FRW health stands (FRW reconciliation chain: sympy ADM reduction and Lean, 11 theorems, standard axioms only; MUTATE 3/3 fail). The P block couples to the lapse only. The static-limit offset is half the earlier value, bounded by $1.1 \times 10^{-10}$. Still open: the AC7 Q2 / AC10–AC15 post-Newtonian chain carries the legacy factor and has not been rederived.

**Solutions** (QX17C, CERTIFIED as a solver). A conservative finite-difference Newton solver with a decaying Bessel outer boundary converges in 1–6 steps on both footings, about 5 s in all. It reproduces the law at $L_\star \to 0$ (worst $3.9\times10^{-7}$), the solar hedgehog to $7.1\times10^{-6}$ of $g_\star = GM/L_\star^2$, and a $10^{10}M_\odot$ galaxy on the law to $4.8\times10^{-6}$; full-GR minus weak-field is $4.1\times10^{-6}$ (Sun) and $5.4\times10^{-7}$ (galaxy). An isolated point Sun reaches only 0.71/0.67 of the law at $10^3 L_\star$, so whether a galaxy built from screened stars shows the law is open.

# 5 Every gate, with its frozen label

The full table, one row per gate, is closure_2026_11_06/CLOSURE_FINAL.md; PAPER69's verdict table is its deposited form and is carried unchanged. The rows below are the 2026-11-06 lanes that came after PAPER69.

| Gate | Label | Lane |
|---|---|---|
| *Equations and health* | | |
| Covariant field equations, v3s and AC2q | CERTIFIED vs direct variation | QX17, QX17A |
| Field equations with the adopted blend stiffness (general $F$) | CERTIFIED with $\Pi F'(\Pi)\, h_{mn}$ in the stress tensor; the $\Pi_+ \to F'$ shortcut alone FAILS the metric equation | QX17D |
| 3+1 form, Lean kernels | Lean PASS; lane NOT CERTIFIED (numerical row, solver) | QX17B |
| Static spherical solver | CERTIFIED, both footings | QX17C |
| $\Pi^6$ repair, curved 3D spherical switch | PASS; adopted by the owner 2026-11-07 as the stiffness; non-radial modes untested | BL3 |
| AC2q decelerating slow roots | HARMLESS, scoped; $\epsilon_\phi = 2$ NOT REACHED by any record history | EP1, EP2 |
| Nonlinear fold, shock-capturing | NO VERDICT as frozen (viscosity check fails); no pair where AC2q regularizes | FS1 |
| Nonlinear fold, viscosity-free reading | DIFFERS, both footings: the 80 clock-only pairs fold 1.2 % earlier (DIFF-TIME EARLIER, edge-carried); the soft and phantom pairs (160) read SAME-REACH; no row is REGULAR and no pair shows AC2q regularizing; all checks pass | FS2 |
| Moving Sun | MS1, MS2: NO VERDICT, bracketed (largest time-mean on any rung 0.147 of the line). MS3: PASS (moving Sun, scoped), FINAL for the door, with no MS4; every cell, gate and bracket below the line, with the largest at 0.617 of it (G1 Mars, M490 REFLECT bracket). The PASS rests on the frozen time-spread classes (NOISE / OSCILLATORY), not on a converged extrapolation. Disclosed: the ultra runs used a different memory allocator setting (allocator only; the chunked solver reproduces bitwise) | MS1, MS2, MS3 |
| Light-cone causality | FAIL (physical, far below data); no repair in the khronometric class | AC18, SV2 |
| *All-gates battery (static gates G1–G8; G9 CMB)* | | |
| Six candidates on G1–G8 | NOTHING PASSES ALL STATIC GATES; every candidate fails the Coma UDGs by at least $3.97\,\sigma_U$ | GB1 |

| Gate | Label | Lane |
|---|---|---|
| Cold fluid: $\mathcal{K}(Q)$ dust, amplitude freed | FAIL: passes the background and CMB lensing, fails G1–G7; cosmic capture 4.111 in every class | KD1 |
| Cold component decaying late | FAIL: CMB-lensing $\chi^2$ 46.3/53.0 against 18.48 at the longest lifetime discs allow | LD1 |
| Hot thermal fermion, 11.1377 eV | FAILS 7 of 9 gates; CMB-lensing $\chi^2$ 166.0–194.2 | HC1 |
| Shock-entropy field (proxy) | FAIL: X-COP and groups; need falls outward while the measured adiabat rises | EN1 |
| Shock-entropy sector in the action | FAIL by the no-tuning rule; static solutions isentropic (f $\leq$ 0.04) | SE1 |
| *The step as several causes* | | |
| Multi-cause step | MULTI-CAUSE IN PART: IMF explains the ellipticals and KiDS | ST1 |
| Joint systematics budget per class | IN PART: ellipticals, KiDS and bright M31 dSphs EXPLAINED; X-COP +0.440, groups +0.232, Coma UDGs +1.120 NOT EXPLAINED (canonical medians); CMB SEPARATE | SE2 |
| Environment, re-read | NO: no environment key passes; satellites need about 1.6× the phantom, Coma UDGs 4–46× | EV1 |
| Disc-only exclusion | every direction DEAD; even a perfect (oracle) exclusion fails GB1 | DX1 |
| One action-native scalar for both walls (OQ2 kept by the owner) | NO VERDICT as frozen. At the frozen fallback $N_{\rm min} = 8$ the S1 reading is DEAD on both footings, every start: SPARC vs groups at $\log q$ 3.0–3.4, gap up to 0.497 / 0.483 dex. Web row YES: the CMB wall needs the law's phantom off on the web ($F_{\rm web,max}$ 0.00481 / 0.00435; with the kernel corrected 0.00798 / 0.00720, Section 6). SEPARATED NO, as frozen | OQ1, OQ2 |
| Outer edge of the law from the switch | NO EDGE IN THE ACTION: for an isolated galaxy the static switch variable is $\Sigma/X = 2+\rho_{\rm ph}/\rho_{\rm DE} \geq 2$, so the phantom halo runs on until an external field ends it (the standard EFE radius, already graded) | outer-edge check |
| *Owner-opened test lanes, 2026-11-07 (adopt nothing)* | | |
| Host-held dust with the measured EFE capping the wells | FAIL (clusters, 0/20 cells, both footings); with no dust the EFE moves the KiDS baseline misfit to +0.870 | QX7 iter_6 |
| Remnant tracing the hot gas | SHAPE FAIL, both footings: the missing mass is more concentrated than the hot gas | QX7 iter_7 |
| Asphericity of hydrostatic masses in the law's field | NEGLIGIBLE ($\leq 0.005$ dex, wrong sign), both footings | QX32 |
| Everywhere matrix (23 rows × 17 columns, every cell sourced) | NO ROW PASSES EVERY COLUMN. The minimal framework cover is three rows that contradict each other (KD1 dust, QX27 retention, a literature row); UDGs, causality with well-posedness, and a derived $a_0$ stay uncovered by every framework row. Two effective fields (BK, QUMOND $\beta$ 2.5) FAIL GB1 G2–G7 on both footings; BK passes only the ellipticals (+0.123 / +0.103) | EW1 |
| Law switched off on the single-stream web (khronon multi-stream key) | FAIL (stage 2, population phantom), both footings. KiDS is kept: all 59 points lie inside the law-on turnaround radius. But the phantom that the galaxy population must keep to hold KiDS sources CMB lensing at $q_{\rm pop}$ 0.936 / 1.079 of the dust target at $z$ 1, against a line of 0.0354 (×26 / ×30; ×11 / ×12 self-consistent). This holds for any key that keeps the law on where KiDS needs it | WL1 |
| Two-switch door (law switch plus charge-switched sector) | graded by WL1: FAIL; EW3 keeps the map row only, no graded number | EW3, WL1 |

| Gate | Label | Lane |
|---|---|---|
| Fresh search for a new local action | no local action found; C1, C2 and C3 all FAIL, both footings. C1, any phantom multiplier of an absolute field length (the first absolute-scale grade): KiDS shift +0.283 / +0.260 against 0.05. C2, phantom screened beyond a second length: the natural local form does not screen, and the web forces that length to 0.3 Mpc or less (1.0 Mpc with the kernel corrected, Section 6), where KiDS fails. C3, two static branches: the static operator is not elliptic (radial), and the groups fail | FA2 |
| Nonlocal, history and global actions | NO CANDIDATE SURVIVES, both footings. I1, causal memory of the phantom: FAIL on the web row (r 75.4 / 78.8; 19.1 / 18.8 with the kernel corrected). I3, formation-epoch $a_0$ memory: FAIL G2–G7 and the web row. I4, conserved phantom charge: FAIL on the web row, NO VERDICT on G1–G7. Disclosed by the lane: I1's MUTATE M1 did not bite as frozen, and peak RSS (1.87 GB) exceeded the frozen 1 GB line | FA3 |
| AeST (v4) on the web, in hot wells and in discs | NO-GO FOR THE CLASS, both footings: one length $1/\mu$ cannot hold both the CMB-lensing amplitude ($\mu \geq 10\,\mathrm{Mpc}^{-1}$; $r$ 1.037 / 1.045 at 10; $\mu \geq 0.63$ / 1.0 with the kernel corrected) and KiDS ($\mu \leq 0.063\,\mathrm{Mpc}^{-1}$); the gap is ×160 (Lean `window_empty`; ×10 / ×16 with the kernel corrected, the window still empty, Section 6). X-COP and groups FAIL at every computed $\mu$: static dust $\leq 0.03\,M_b$ against a need of 2.39 / 2.07 $M_b$. The beyond-hydrostatic gate fails (0 % of X-COP radii). Primary CMB FAIL (read). At host-compatible $\mu$, SPARC, Cassini, $c_T$ and lensing = dynamics PASS. Two missed known answers are stated by the lane; no label moves | EW2 |
| Late-formed compact baryonic objects ($10^2$–$10^7\,M_\odot$) as the hot-system half; the CMB wall untouched | FAILS (rows), both footings: X-COP shape, groups shape, Bullet. The budget is not excluded (lower bound 0.054 / 0.050 of the BBN baryons). No single formation rule: every key is S1 DEAD (gaps 1.50–2.53 dex). The dynamics window shrinks to $10^2$–$1.8 \times 10^2\,M_\odot$. Objects tracing the stars miss X-COP's shape (rms 0.1208 / 0.1138), while an HSE-traced profile passes, so the needed profile lies between the gas and the stars. Groups pass 11 / 9 of 20 (canonical) and 10 / 8 (alt) against 16 needed. Disclosed by the lane: the Bullet child process peaked at 1188 MB, over the 1 GB line | NA2 |
| Last two FA1 doors: disformal bulk-motion selector (I5), defect network of the $a_0$ scalar (I7) | I5 DEAD BY REDUCTION, both footings: every scalar the disformal metric allows reduces to dead keys ($\gamma$, $X$, $\rho$, $p$); with matter on the disformal metric, GW170817 confines $BX$ to $[-6.2 \times 10^{-15}, +1.3 \times 10^{-15}]$, so the selector varies by at most $6.8 \times 10^{-20}$; no monotone weight passes S1; web lensing $r$ 72–86 against 1.072 (27–39 with the kernel corrected). I7 DEAD, both footings: with $\eta$ set by $a_0$ (2.240 / 2.463 meV) the network forms only at $z$ of about 9; with $\eta$ free, scaling networks track the background ($w$ 0.167 at $z$ 3400); the network is at most $3.7{\times}10^{-4}$ of the baryons at $z$ 3400 and at most $1.3{\times}10^{-8}$ in clusters; gauge monopoles are particle-like. Disclosed by the lane: KA7 restated in amendment A1 before the rerun (labels unchanged); MUTATE M2 and M4 bite only weakly; the first run peaked at 1.025 GB, the rerun at 0.858 GB | NA1 |

| Gate | Label | Lane |
|---|---|---|
| The $a_0$ law plus an 11.1377 eV thermal fermion (test lane; the no-particle rule relaxed for this lane only) | does NOT pass every EW1 column, both footings. Run as a consistency test it reproduces the literature: X-COP 6/7 consistent, $M_{\rm req}/M_b$ 1.8–3.4, using 32–62 % of the supply, phase space never binding; run as a prediction (HC1) X-COP fails. Groups FAIL (14/20 / 11/20 against 16; the law over-predicts the shell); ellipticals FAIL 0/7 on phase space; AGC 114905 and Malin 1 FAIL for any fermion; KiDS G6 −0.098 dex FAIL; primary CMB $\Delta\chi^2$ +2716 / +2599 FAIL; CMB lensing, Coma UDGs and the colour split FAIL. SPARC PASS (not robust); Cassini and $c_T$ PASS only with v3s/AC2q as the law's action. Disclosed by the lane: MUTATE M1 (mass halved) has NO POWER on X-COP, and the X-COP loader holds 7 clusters, not 12 | EW4 |
| KiDS isolated-lens kill of the late-heated window (QX7 iter_4), as a joint systematic budget | NO VERDICT as frozen (the 4× MUTATE cannot bite on cell C3); no cell softens, both footings. KILL HOLDS on C2 and C4, NO POWER on C1 and C3. The hot gas measured by the kinetic SZ alone accounts for the raw +0.15–0.20 dex excess at 100–300 kpc; the posterior excess of data over the law is −0.21 to +0.05 dex (canonical) and −0.25 to +0.01 (alt), and the implied line is 0.05–0.24 dex (canonical) and 0.00–0.20 (alt), against the window's 0.17–0.25. Disclosed: the lowest mass bin uses an extrapolated gas row, and canonical C4's hold rests on it | KX1 |
| SE2's two fragile rows, re-graded | Coma UDGs NOT A TEST (2 of 9 admissible at pericentre $d/3$; no orbit on disk); groups NOT EXPLAINED (+0.259 / +0.221), fragile; the residue is X-COP + groups + CMB lensing | UA1 |
| Late-heated window on the cluster gates (QX7 iter_4 cells C1–C4) | FAIL on every cell, C3 included, both footings. Every cell passed CMB lensing as frozen ($\chi^2$ 8.70–11.69; with the kernel corrected that gate reads NO VERDICT, Section 6), $f\sigma_8$ at $z < 1$ ($R_f$ 1.000) and the forest at $z$ 2–4, and fails X-COP's profile, the groups and the Bullet. C3: X-COP +0.324 / +0.287 (inner bin +0.48: what evaporates from galaxies evaporates from clusters too), groups 0–1 of 20, Bullet FAIL (the sector is a fluid at Mach 1.7–1.9, ram-stripped, and follows the gas). Disclosed: the KX1 narrowing to C3 arrived after the numbers (C3 was frozen and graded as frozen); the Bullet child peaked at 1.48 GB, over the 1 GB line | CE2 |
| Residual surface: does any pair of measured variables carry the law's missing mass across classes? | NO VERDICT as frozen: the Newtonian-$f$ MUTATE does not bite, because $g_{\rm obs}$ was on the frozen variable list (from the brief) and makes the law-off $f$ an exact function of $(g_N, g_{\rm obs})$. Main reading NO PAIR, both UDG readings. The closest pair, $g_N + g_{\rm obs}$ ($\chi^2/N$ 0.433 / 0.440), is circular and fails canonical leave-one-out. The best pair without $g_{\rm obs}$, depth + $\sigma$ ($\chi^2/N$ 1.483 / 1.434), leaves SPARC at $z$ −4.49 / −3.87, X-COP at +3.82 / +3.48 and the UDGs at +3.46 / +3.48; without $g_{\rm obs}$ no pair passes leave-one-out after dropping any class. SPARC is GB1's 145 galaxies | RS3 |
| *The kicked carrier, re-graded (RESOLUTION, 2026-11-07)* | | |
| Does the action release a kick when its switch changes branch? | NO RELEASE TERM, both footings: no isolated host changes branch at virialisation or at the acceleration crossover (min $\rho_{\rm ph}/\rho_{\rm DE}$ +0.000507 / +0.000462), and the blend gives zero latent energy at the branch point. Any kick is a postulate, not the action's | CK2 |
| September's kicked collisionless carrier at today's needs, $v_k$ 500–650 km/s | FAILS (rows) at every $v_k$ on both footings; the binding gate is the SPARC high-mass tail, where every galaxy with $v_{\rm flat} \geq$ 250 km/s keeps the carrier. The kick $v_k$ is hand-set | CK1 |
| Cosmology under the kicked carrier | NO VERDICT (the CMB-lensing gate has no power); among gates with power, $v_k$ 550 passes and 575 / 600 fail the forest | CK4 |

| Gate | Label | Lane |
|---|---|---|
| A cold component born clumped at the future nodes, never in galaxies | FAIL, both footings, every clump mass: the primary CMB tolerates a bias of only 0.99670–1.00142, while a peaks-born population carries 45–1866 (bias 1.2 / 1.5 / 2.0 give $\Delta$ +4.10e4 / +5.57e5 / +6.35e6); the forest allows at most 0.13–0.28 % of the component in clumps; the clumps' own Poisson power fails CMB lensing at $10^{14}$ / $10^{15}\,M_\odot$ (+75.9 / +239.8) and passes at $10^{13}$ | NC2 |
| Do cluster-member early types carry the component? | UNRESOLVED by rule, both footings: members sit +0.011 [−0.044, +0.065] / +0.012 [−0.043, +0.068] dex from field early types at fixed stellar mass and dispersion, so at most about 16 % extra mass at $r_{\rm half}$; stripping cannot hide a residue; the KiDS isolated-lens control is clean; CLASH's lensing-only slope leans to an early clump's sign with every interval containing 0 | NC3 |
| The born-clumped clump spectrum (a record row; NC2 closed the candidate) | FAILS (rows) on both footings, binding K3: hosts inside $R_{500}$ hold 0.0133 of the cosmic baryons and so 0.0091 / 0.0078 of the cold component (0.025 / 0.022 in the most generous bracket); clumps would need 110 / 128 times the observed host abundance (40 / 45 generous); even the ceiling with every baryon in a host is 0.75 / 0.68. K5 also fails: the top-hat collapse in the law's field fits X-COP at rms 0.166 / 0.143 at best. A fitted mass trend passes (slope 0.6–2.3 / 0.8–2.2 per dex). Disclosed: the Schmidt test never passed ($\langle V/V_{\rm max}\rangle$ 0.403–0.465), and a factor 2–3 would still leave the shortfall $\geq 13$ | NC4 |
| Is the CMB's cold component independent evidence? (F1, audited with own likelihood code) | F1 CONFIRMED, both footings: baryons alone fail the primary CMB by $\Delta$ +172,915.6 / +166,674.4; the cold fit is adequate ($\Delta_{\rm cal}$ −65.21 / −65.07 against $\chi^2_{\rm cal}$ 3805.37) at $\omega_d/\omega_b$ 5.410 / 5.412 (0.973 of 5.562); an own Newtonian-gauge code reproduces that the khronon clock cannot cluster ($Q \leq 3.4\times10^{-11}$ at $z$ 1100). Disclosed: CAMB is the Boltzmann code, as in QX15 (CLASS differs by 3.8); the baryons-only fits sit at bracket edges; one check peaked at 1.81 GB for 10 s | F1A |
| Can the action's own dust term ($\mathcal{K}(Q)$, at KD1's amount) form the cluster core? | CANNOT FORM CORE, both footings, for every $\mu_K$ in 0.1–$10^4$ Mpc$^{-1}$: in the law's $1/r$ field the hydrostatic dust falls like $\ln(r_s/r)$ with a hard surface, so one Jeans length sets both core and extent. A 50–100 kpc core comes with a surface near 160 kpc (concentration 1.000, inner bin over by 1.1 dex); a concentration near 0.25 needs a 360–400 kpc core; the best per-bin rms anywhere is 0.226 / 0.222. The closest profile ($\mu_K$ 19.05) is $4.67\times10^5$ below the BBN bound, and galaxy capture stays at 4.111 / 4.112. A planted cored isothermal passes the same machinery (rms 0.091) | KQ1 |
| Does the fold machinery show a release layer at 500–650 km/s? (the outside solver's kick identity, $v_k = 2c(a_0L_\star/c^2)^{1/3}$ = 577.6 / 571.3 km/s) | NO VERDICT (check failed): the planted 577.6 km/s control did not read back in the band (it regularizes late and reads 317.7 / 309.2 km/s). The graded rows read $v_{\rm post}$ 0.024–0.157 km/s and $R_\delta$ $5\times10^{-10}$–$3\times10^{-7}$, all 160 readings outside the bands; the speed tracks the infall ($d\ln V/d\ln v_f$ 1.000 / 0.95) and doubling $L_\star$ leaves it unchanged (ratio 1.000, where the identity predicts 1.260). Disclosed: the brief's $\lambda_{\rm crit}$ was off by a factor 10 (22.53 / 24.04 kpc, not 2.09 / 2.25); the reduction is non-relativistic ($1-X/X_{\rm bg} \leq 2.4\times10^{-7}$), so the identity's relativistic regime cannot appear | AK1 |

*The CMB requirement, anatomised, and two owner-named lanes (2026-11-08)*

| Gate | Label | Lane |
|---|---|---|
| Which features of the primary CMB demand the cold component? (anatomy of F1A's baryons-only $\Delta$) | NO VERDICT as frozen: the M1 swap moves H1/H2 by $+343\,\sigma$ and H3/H2 by $+197\,\sigma$ in the same direction, ratio 0.57 inside [0.5, 2] (the third peak is a compression peak too). Block C CLUSTERING PINNED, both footings. The baryons-only deficit is spread over TT, TE and EE and every multipole range; no piece carries more than 19 %. A smooth matter-like density ($\omega_X$ 0.108, about 5× the baryons) that sets equality and the expansion but does not cluster recovers 79 % ($\Delta_E$ +35,812 / +35,378); raising $\omega_b$ alone recovers none. The rest needs still potential wells through $z$ 1100: without clustering H3/H2 misses by $-36.8\,\sigma$, H1/H2 and the first peak by $-16\,\sigma$; the cold fit meets every feature within $1.1\,\sigma$ | CMB1 |
| Before recombination, does any degree of freedom other than the clustering dust supply a still Weyl potential ($\geq$ 0.5 of the dust's at $k$ 0.03–0.3 $\mathrm{Mpc}^{-1}$)? | NO, both footings. The khronon capped at the galaxy allowance: still share 0.043 / 0.024; the clock about $10^{-20}$. The P sector has no propagating mode (Hessian rank 0). The matter equations carry no khronon field, $c_s^2 = 1/(3(1+R))$ exactly; the slip is at most $7 \times 10^{-4}$ and follows the source. The law at $a_0(z)$ 1–100 times today's: its phantom follows its source ($r$ 0.992–0.997), still share at most 0.28 / 0.30. Open, stated: in a toy oscillator a following phantom at 30 times or more raises the odd peaks (H3/H2 1.30 / 1.36 at 100 times, against the dust's 1.447); that needs $\rho_{\mathrm{DE}}$ about 900–$10^4$ times today's at $z$ 1100, and no history in the record gives it | CMB2 |
| F1A's G4 starting state (amendment A2, prompted by CMB2) | G4's printed contrasts depend on the starting time; a start consistent with the Poisson constraint converges (−26.65 / −94.13 at $k$ 0.05 / 0.3); $Q_{X1} \leq 3.7 \times 10^{-12}$; G4 REPRODUCED, F1 CONFIRMED unchanged | F1A A2 |
| Is there one vacuum-response length in the action whose identity fixes $\kappa = \frac{1}{2}$? | NO IDENTITY, both footings. The khronon kinematics carry the 2 exactly ($\nabla \ln X = -2A/c^2$), but setting the response length at $a_0$ equal to a vacuum length gives $\kappa^2 = 1/(4q)$, which is $\frac{1}{2}$ only for $q = 1$, the $\pi$-free length $c/\sqrt{G\rho}$ already posited (KF1's $\xi = 1$). The six GR lengths give 1.447, 1.772, 0.921, 0.282, 2.507, 0.230; the cancellation identity has no root; equipartition gives 4.525 / 6.399; the blend identity depends on host mass. The vacuum's response carries at most $6.9 \times 10^{-8}$ of the cluster need. Lean: 7 theorems, MUTATE twin fails 4/4. $\kappa$ stays fitted | VR1 |
| Settling-age dependence of the action's dust (a reduction; no criteria, no script, no number) | NO TERM IN ACTION: $\mathcal{K}$ depends on $Q$ alone and the field equations are local in time. Every route by which a settling age could act (capture, hot early, born clumped, kick, late heating, self-expulsion, formation-epoch and merger keys, local keys, a perfect oracle) is a closed cell on both footings, and the Coma UDGs, old and settled, carry the largest need in the record | SA1 |
| The $a_0$ law with $\kappa$ free plus the CMB's cold component (quarantined test lane: the owner lifted the no-cold-component rule for HY1 only) | NO VERDICT / NO VERDICT on the frozen M3 rule: doubling the dust worsens discs and cluster cores together, so the control cannot bite (a gap in the control's design, disclosed). Under it, FAIL on both footings. The record's own dust dynamics (KD1 model P at the one $\mu_K$ BBN allows) put the component in every galaxy and cluster as a central lump of about 4.1 times the baryons: SPARC rms 0.465–0.658 / 0.464–0.680 against 0.110 at every $\kappa$, and X-COP cores overfilled by about 1.1 dex at every $\kappa$ and any amount. A weaker law ($\kappa$ in [0.05, 0.17] / [0.05, 0.13]) holds KiDS, the group step and the Solar System, but only if discs shed at least 96 % of the dust, which is the second fact again. Bullet and the law on the linear web NOT EVALUABLE. Adopts nothing | HY1 |

| Gate | Label | Lane |
|---|---|---|
| Assumption audit of the inference that the peaks need a cold component of about 5.4 times the baryons (A1 law inert at $z$ 1100, A2 lensing, A3 recombination and $N_{\rm eff}$, A4 initial conditions, A5 the identification with galaxy dark mass, A6 nuisance templates) | RUNNING (criteria frozen e46081b7a9; assumption audit A1–A6; result in v1.6) | CMB3 |
| Rows touched by the kernel erratum | the CMB-lensing and web-lensing numbers of KD1, SE1, CE2, NA1, FA2, FA3, EW2, OQ1 / OQ2 and SV5 above were computed with the Limber kernel corrected in Section 6; the corrected values are listed there and, by the owner's decision of 2026-11-07, are the standing labels; every row above keeps its FAIL or NO-GO | GL1, GL3, GL4 |

# 6 Erratum to v1.0: the CMB-lensing Limber kernel

The Limber integrator shared by the sector and web lanes (QX3 iter2, `limber_ratio`) omitted the lensing potential's $k^{-4}$. It weighted each redshift by an extra $\chi^{-4}$, so the ratio $S(L)$ was dominated by $z \sim 0.02$–$0.1$. CK4 found it, the code was read independently, and GL2 CONFIRMED it with its own code against CLASS: with $k^{-4}$ restored the Limber shape over CLASS's own spectrum is 0.98–1.006 at $L$ 20–400, against 0.045–33 without it. GL1 (sector side) and GL3 (web side) re-ran every affected lane with the kernel corrected; re-inserting the error reproduces the committed numbers (to $\leq 4.5 \times 10^{-16}$ on the web side), and a wrong power ($k^{-2}$) is the mutation. WG1–WG6, LD1 and HC1 used separate pipelines and are untouched. **The owner accepted the corrected values as the standing labels on 2026-11-07 (closure_2026_11_07/OWNER_DECISION_2026-11-07_ACCEPT_CORRECTED_KERNEL_LABELS.md). Old and corrected values stay side by side below, and the frozen RESULT files are not edited.**

| Label (lane) | Old | Corrected | Reading |
|---|---|---|---|
| Wall 1, CMB side (KD1 G9a, SE1 arm (a)) | $\mu_J$ $2.51 \times 10^4$ | $\mu_J$ 631; the binding side is BBN ($\mu_B$ $8.89 \times 10^6$, unchanged) | HOLDS |
| QX3 iter2 (no window) | $\mu_{\rm min}$ $3.98 \times 10^4$ vs $\mu_{\rm max}$ 79.43 | $\mu_{\rm min}$ 631, still above 79.43 | HOLDS |
| QX3 iter3 (class no-go, 35 cells) | best $\Delta\chi^2$ 92.59, 0/35 | best 2.78, 1/35 ($\beta$ 0.80, $\mu_0$ 1259) | re-stated (owner 2026-11-07) |
| QX7 iter_3 (no window, 31 cells) | closest 17.87, 0/31 | 31/31 inside the lensing line | re-stated (owner 2026-11-07) |
| QX7 iter_4 (FAIL on clusters) | window 270 cells, clusters 0 | window 857 / 857, clusters 0 / 857 | HOLDS |
| QX10 iter_1 (NO VERDICT) | J1 $\Delta\chi^2_{TT}$ 4.30 / 1.05 / 77.2 / 3.47 | $\leq 10^{-7}$ on all four cells | re-stated (owner 2026-11-07) (PASS by its own rule) |
| CE2, CMB-lensing gate | PASS, $\chi^2$ 8.70–11.69 | $\Delta\chi^2$ 0.000 on every cell; its mutation no longer bites | gate re-stated (owner 2026-11-07) (NO VERDICT); CE2's FAIL on X-COP, groups and the Bullet is kernel-free and HOLDS |
| CK4, CMB-lensing gate | NO VERDICT | NO VERDICT | HOLDS |
| SV5 iter_1, web amplitude W1 (line 1.072) | $r$ 74.74 / 89.36 | 32.53 / 38.66 | HOLDS (FAIL) |
| SV5 iter_1, web shape W2 | $\Delta\chi^2$ 0.019, PASS | 33.5 / 34.1, FAIL | re-stated (owner 2026-11-07) (stricter) |
| SV5 T0.2, latest $z$ for a late Jeans length of 30 / 100 / 1000 Mpc | none / none / none | 2.0 / 1.5 / 0.75 | re-stated (owner 2026-11-07) |

| Label (lane) | Old | Corrected | Reading |
|---|---|---|---|
| SV5 iter_3, $a_0(z)$-tied Jeans scale, H-native | $M^*$ $\Delta\chi^2$ 876.7, FAIL | 0.016; K-mono 0.52 | re-stated (owner 2026-11-07) (WINDOW-CONDITIONAL; growth and lensing gates only) |
| WL1, population phantom line | $q_{\rm pop}$ 0.936 / 1.079 vs 0.0354 | identical (kernel-free) | HOLDS |
| NA1 I5, web $r$ | 72.2–89.4 | 27.4–38.7 | HOLDS (FAIL) |
| FA3 I1, web $r$ | 75.39 / 78.77 | 19.12 / 18.75 | HOLDS (FAIL) |
| FA2 C2, web cut-off $L_{\rm max}$ | 0.3 Mpc | 1.0 Mpc; KiDS still fails there ($-0.074$) | HOLDS |
| EW2, web window | $\mu_w$ 10, gap $\times$160 | $\mu_w$ 0.63 / 1.0, gap $\times$10 / $\times$16 | HOLDS (window empty) |
| OQ1 / OQ2, web row | $F_{\rm web,max}$ 0.00481 / 0.00435 | 0.00798 / 0.00720 | HOLDS (YES) |
| QX28 G6b | $\Delta\chi^2$ 140.1 / 70.0, FAIL | 0.006 / 0.004, PASS | re-stated (owner 2026-11-07); G6 still FAILS on the kernel-free G6a |
| QX27, QX29, QX31 | FAIL; PASS (A-web conditional); ROBUST | labels unchanged | HOLDS |
| QX15 cells C1 / C2 / C4 (found by F1A) | $\Delta_{\rm cell}$ +0.66 / +0.27 / +0.57, PASS | $+1.1\times10^{-5}$ / $+6.5\times10^{-6}$ / $+1.1\times10^{-5}$, PASS | HOLDS |
| QX15 cell C3 | $\Delta_{\rm cell}$ +5.95 / +6.03, TENSION | $+4.1\times10^{-5}$ / $+4.2\times10^{-5}$, PASS | re-stated (owner 2026-11-07) |
| QX15 P1 power control | +192.46 / +189.91, bites | kernel-free, reproduced; bites | HOLDS; QX15's main fits (F1) do not use the kernel |

What moves is late, small-scale suppression: with the kernel corrected, PR4's shape at $L \leq 400$ is nearly blind to it. No move opens a door by itself, because the cluster, group and KiDS rows that close those doors are kernel-free. Wall 1 holds on its BBN side; on the CMB side alone the galaxy-versus-lensing gap narrows from $\times$316 to $\times$7.9.

# 7 The CMB requirement, anatomised

CMB1 asks what the primary CMB's demand for a cold component is made of; CMB2 asks whether anything in the action other than the clustering dust can meet it. Both run on the framework background, with no ΛCDM input, on both footings.

1. The deficit of baryons alone ($\Delta$ +172,915.6 / +166,674.4, F1A) is spread over TT, TE and EE and every multipole range; no piece carries more than 19 %. Raising the baryon density never helps.
2. About four fifths is the expansion history. A smooth matter-like density of about five times the baryons, which sets matter–radiation equality and the distance to $z$ 1100 but never clusters, recovers 79 %.
3. The last fifth ($\Delta_E$ +35,812 / +35,378) is clustering. The third-to-second and odd/even peak heights and the first-peak position need potential wells that stay still through recombination, made by a component that does not feel photon pressure.
4. Nothing in the action except the clustering dust supplies such wells (CMB2). The khronon, capped at what galaxies allow, gives 0.043 / 0.024 of the dust's still potential; the P sector has no mode; the preferred frame does not enter the photon–baryon equations; and the law's phantom follows its source.
5. The law is inert at $z$ 1100 on the record's own numbers. With $\rho_{\rm DE}$ constant its peak ratios are those of baryons alone (H3/H2 1.008 vs 1.006). The one open corner, $a_0$ at $z$ 1100 thirty or more times today's, needs a dark-energy history 900–$10^4$ times today's at $z$ 1100; even such a history leaves the action inert (X35, below).

So the peak inference holds on the framework's own background: at recombination there is a pressureless, clustering component of about 5.4 times the baryons, and no term of the action other than its dust supplies it. What the record breaks is not that inference. It is the identification of this component with the agent that flattens galaxy rotation curves: galaxies hold almost none of it today, and bound hosts inside $R_{500}$ hold about 1 % of it (NC4).

CMB3, an audit of the inference's assumptions A1–A6, is running at this version and reports in v1.6. The parts of A1 and A3 that concern the action are already answered by CMB2: the law's phantom follows its source (D4), and the matter equations carry no khronon field (D3). Their own numerical tests and A2, A4, A5 and A6 are in flight. CMB2's open corner is already answered in the record by X35 (rebuild_2026_10_22/X35_a0_tracks_ede; UNDECIDED by its frozen rule). Early dark energy does put $a_0$ at $z$ 1100 at 1,147–7,565 times today's, but in the covariant action the extra potential is the weight of khronon charge, which stays identically zero without a separate cold component, so the potential is the GR potential of baryons, photons and neutrinos for every $a_0$ (D1 NULL; a nonlinear 1-D solve from a non-zero start returns $\max|\Xi| \; 4.9 \times 10^{-29}$). The odd-peak rise of CMB2's toy appears only in the static AQUAL reading, which is not the action, and that reading makes the photon–baryon fluid Jeans-unstable at first-peak scales ($k_{\rm J}$ 0.024–0.039 $\mathrm{Mpc}^{-1}$). The full Boltzmann check CMB2 named is therefore not owed. Early dark energy is not neutral late, though: through the sound horizon it moves CFG689's native $\sqrt{X}$ at $z$ 2.33 from 1.00 to 1.49 (a flat-$\Lambda$ estimate).

# 8 What an all-gates ingredient would have to supply

On top of the law, the extra gravitating mass as a fraction $f$ of the baryons (KD1 diagnostic, canonical / alt):

| Where | Needed $f$ |
|---|---|
| SPARC discs | $\leq 0.33$ / 0.19 |
| X-COP clusters | $\sim 4.2$ / 3.8 |
| groups | 1.2–3.5 / 0.9–2.9 |
| X-ray ellipticals | $\sim 0.9$ / 0.8 (about half of it IMF) |
| dSph satellites | $\sim 1.7$ / 1.3 |
| Coma UDGs | $\sim 80$ |
| the CMB | cold at $z$ 3400 ($w \leq 0.01$), lensing $\chi^2 \leq 18.5$ at $L \leq 400$ |

# 9 Four walls

1. **Cold fluid** (KD1, QX3, QX7). Every well captures it, discs included. The CMB and BBN need $\mu_K \geq 8.9 \times 10^6\,\mathrm{Mpc}^{-1}$; galaxies allow at most 79.
2. **Warm fixed-temperature sea** (QX12). Retention $\exp(\mathrm{depth}/c_s^2)$ is too sharp; the temperature would have to track each host.
3. **Hot thermal fermion** (HC1, HF1, VW1). Hot enough to stay out of discs, it under-fills clusters, cannot fill dSphs or UDGs, and fails CMB lensing; cold enough for the CMB, discs capture it. No window.
4. **Shock-entropy sector** (SE1). A conserved, non-cooling sector heated by its own shocks either sits isentropic, and holds almost nothing, or needs a different entropy history for each class at the same entropy. Its conserved rest mass also caps a host near the cosmic share, against the $\sim 80$ the Coma UDGs need.

The common reason: an ingredient that tracks each host's virial state has to virialise with the host; if it is also cold early, nothing keeps it out of discs. And the CMB needs the extra gravity at $z \simeq 3400$, where nothing is virialised, so any key that is on only in today's hot systems is off exactly where the CMB needs it on.

# 10 The residue: the law's measured boundary

SE2 draws every systematic knob jointly per class (IMF, velocity-dispersion scale, distance, anisotropy, membership, 3-D position, tidal admissibility). The posterior step of the plain law, in dex (median, 95 % interval):

| Class | canonical | alt | label |
|---|---|---|---|
| X-COP clusters | +0.440 [+0.372, +0.484] | +0.408 [+0.341, +0.451] | NOT EXPLAINED |
| groups | +0.232 [+0.078, +0.362] | +0.194 [+0.041, +0.324] | NOT EXPLAINED |
| Coma UDGs (9) | +1.120 [+0.541, +1.718] | +1.075 [+0.472, +1.673] | NOT EXPLAINED |
| X-ray ellipticals | +0.067 | +0.050 | EXPLAINED |
| KiDS early/late | +0.085 | +0.084 | EXPLAINED |
| bright M31 dSphs (21) | +0.228 | +0.219 | EXPLAINED |
| CMB-lensing shape | cold extra gravity at $z$ 3400; no key passes the hot classes | | SEPARATE |

The groups row is the weakest ($z$ 3.04 / 2.55). Gas in the Coma UDGs is not in the budget and would lower their step; hydrostatic bias is set to 0 and would raise the cluster and group steps. These numbers are the boundary of the law as measured, not a mechanism. One profile meets both the inner-core requirement and the gas-to-stars band: a cored isothermal with $r_c$ 50–100 kpc passes NA2's per-bin rule for 7/7 clusters on both footings ($r_c$ 100 kpc: rms 0.086 / 0.086), though no collapse model in the record produces it (NC4, reported).

**Update (UA1).** UA1 re-graded SE2's two fragile rows. *Coma UDGs: NOT A TEST* on both footings: with tidal admissibility taken at pericentre $d_{\rm mean}/3$ in the Coma field, 2 of 9 are admissible. No orbit or proper motion exists on disk for any of the 9, so the label rests on the $d/3$ convention; with pericentres at $d/2$ or wider, 5 or more would be graded (not computed). NOT A TEST says the class cannot show it is in equilibrium; it does not say systematics explain the step. *Groups: NOT EXPLAINED* under the full single-width budget, +0.259 [+0.052, +0.459] (canonical) and +0.221 [+0.009, +0.415] (alt). The non-thermal term raises the step; distance and clumping lower it by about 0.01 dex each; the cross-catalogue term is 0.064 dex; the Eckmiller group masses are PROVISIONAL. The row is fragile: the alt lower edge is +0.009, and its sign rests on the non-thermal term (the inverted sign drops the median by 0.088 / 0.087 dex). With the same cut the bright M31 dSphs stay EXPLAINED (15 of 25). **The residue is X-COP + groups + CMB lensing**, the group-and-cluster scale, not the cluster scale alone.

## 11 Kill tests, frozen

Three tests can kill the law with data not yet in hand. Each is frozen before the data; nothing else is pending.

1. **Gaia DR4 wide binaries: Newton.** v3s screens binaries, so it predicts Newtonian motion: the frozen pipeline returns $\gamma_\infty$ 1.0019 / 1.0100 (canonical / alt), not separable from Newton and separable from plain QUMOND (1.1494 / 1.2081). The prediction is sealed (CQ2, SHA-256 1d65509dbf). Scored as the screened (medium-first) reading of the frozen DR4 rows (owner decision 413c1c8e24), v3s is excluded iff $\hat\gamma - 1 \geq 3\sigma$, with the gate $\sigma_{\rm sys} < 0.048$ / 0.058 and the R rule (KILL iff $R - 1 \geq \max(0.01, 3\sigma_R)$). At the registered error (N 30,000, $\sigma_{\rm sys}$ 0.020) v3s is EXCLUDED for $\hat\gamma$ in 1.083–1.23; at the realistic N of about 8,000 the edge moves up.
2. $a_0(z)$ **flat to** $z$ **2.33.** $a_0$ tracks $\rho_{\rm DE}$ (CFG689 quantiles, BAO + SN, no ΛCDM input). No sample on disk can decide it: all 52 cells are BLOCKED, because each sample's common mode (0.30–0.60 dex) is 4–11× the largest $e_{3\sigma}$ (0.084 dex at $z^*$ 2.33). It needs a proposal-class ALMA sample with a same-estimator $z < 0.1$ reference (AT2). It kills the law if, at a powered $z^*$, the rival is favoured or neither model fits (AT1's rule).
3. **Lensing = dynamics.** v3s predicts $M_{\rm lens}/M_{\rm dyn} = 1$ to $\leq 6.5 \times 10^{-5}$ at 310 kpc. KiDS isolated lenses give $A = 1.129 \pm 0.206$ (canonical) and $1.035 \pm 0.187$ (alt), PASS but with weak power (V3W W3) The kill line is the frozen grade: FAIL if the KiDS ratio sits more than $2\sigma_{\rm tot}$ from 1, or the matched-host ratio more than $3\sigma_{\rm tot}$.

The exact statistics and kill lines are collected in closure_2026_11_07/PREDICTIONS_2026_11_07.md (52d4f27840), checked by its own script.

## 12 Reading

Every physics class on the record now carries a label, and none passes every gate. The law, and the action within its scope, describe cold rotating discs and the Solar System. The ellipticals, KiDS and the bright M31 dwarfs are accounted for by measured systematics. X-COP and the groups are not; the Coma UDGs are NOT A TEST on the record's orbit convention; and the CMB-lensing wall stands on its own. The elimination argument

(closure_2026_11_07/DEDUCTION_2026_11_07.md) runs as follows. Three facts are measured: something cold gravitates at recombination; galaxies hold almost none of it today; clusters hold it today. No time-independent ingredient carries all three, and every switch on the law is closed. A component whose state changes in time leaves two cells. The escape cell, a late kick above the galaxy escape speed, is closed by X-COP. The puff cell, a late heating that leaves a diffuse halo of about $4\,M_b$ around galaxies, is closed by KiDS once the hot gas is measured rather than assumed (KX1). Its evaporated-core form, C3, fails X-COP, the groups and the Bullet (CE2), so every form of the puff cell is closed. The escape cell, re-graded at today's needs, fails on every kick from 500 to 650 km/s (CK1), and the action supplies no kick, neither in its statics (CK2) nor in its fold dynamics, where the release speed tracks the well and not the identity's 577.6 km/s (AK1). A component born clumped at the nodes fails the CMB's bias tolerance and the forest (NC2), and cluster-member galaxies are as clean as field galaxies (NC3), so the exhaustion covers initial conditions as well as mechanisms (DEDUCTION section 8). One structural fact frames all of this: every host inside $R_{500}$ together holds about 1 % of the cold component the CMB needs (0.0091 / 0.0078; at most 0.025 / 0.022 in the most generous bracket), so nearly all of it, whatever it is, lies outside bound objects (NC4). With F1 now audited by independent code (F1A), all three measured facts rest on independent checks (DEDUCTION section 9); the action deposited here is the full action the record can write, and its one unresolved failure is a contradiction rather than a gap: its own cold dust, forced by the CMB and present in clusters, falls into galaxies where no measurement finds it, and no mechanism or initial condition in reach of the action prevents that. The action's own dust term also cannot form the cored cluster profile (KQ1; DEDUCTION section 10), so the residue's shape is a target no term on the table reproduces. On 2026-11-07 the owner accepted the corrected-kernel labels as standing (Section 6); none of them opens a door, DEDUCTION sections 7–10 are unchanged, and no row in the record is pending. The elimination is complete for every ingredient the record can build. The $a_0$ law with the adopted v3s action is the law of cold isolated gravity; the CMB's cold component, the cluster and group residue, and the absence of either from galaxies are three measured facts that no single ingredient in reach of the action connects. Which of the three gives way is not decidable on disk: the galaxy leg is now the firmest (KX1); the cluster leg rests on seven X-COP clusters and provisional group masses (UA1); the CMB leg is a fit under the footing background, not independent evidence (QX15). The hybrid route, the law plus a component that galaxies hold, is now graded under the record's own dust dynamics (HY1, a quarantined test lane) and fails on both sides at once: the dust over-fills discs and cluster cores together. The owner's lift of the no-cold-component rule covered HY1 only. This is the best remaining answer the record supports, not a closed theory.

## 13 What is unified, and what is not

From closure_2026_11_07/UNIFICATION_STATEMENT_2026_11_07.md, section 3, verbatim:

> Gravity carries one extra scale, the acceleration of the dark energy, a0 = (1/2) c sqrt(G rho_DE), and one action, the screened khronon action with its adopted stiffness, realises it as a complete and certified law of cold isolated gravity from the Solar System to galaxy lensing and gravitational waves. The same relation predicts a constant a0 to z 2.33 and Newtonian wide binaries, which are its kill tests. Beyond the cold isolated regime the universe contains a cold gravitating component that the CMB requires, clusters hold and galaxies lack, and no term in reach of the action, nor any ingredient the record could build, unifies that component with the law; the one candidate that does so adds a postulate under test today. That is the whole of the unification the record supports: one scale, one action, one measured boundary, and a named missing identity.

Since that statement was written, the candidate it names has been graded: CK1 FAILS it on every kick, and CK2 finds no term in the action that releases one. The kernel erratum (Section 6) leaves wall 1 and every web row standing. $\kappa = \frac{1}{2}$ stays fitted and not derivable. VR1 tested the vacuum-response identity a principle would have to supply and finds none: the missing premise is that the vacuum's response length at the switch is $c/\sqrt{G\rho_{\rm DE}}$, and nothing in reach supplies a $\pi$-free length. SA1 reduces a settling-age dependence of the action's dust to closed cells. CMB1 and CMB2 (Section 7) leave the cold component's requirement as measured and its carrier outside the action.

## 14 What waits on the owner

The PAPER68 erratum for the Legendre form. The $\Pi^6$ stiffness repair, whose case for and against was written for the owner (PX1, closure_2026_11_07/PI6_BLEND_ADOPTION_CASE_2026_11_07.md), was adopted by the owner on 2026-11-07 (closure_2026_11_07/OWNER_DECISION_2026-11-07_ADOPT_PI6_BLEND.md). Every owner-opened test lane above has landed with its label, and none is still running; none adopts anything.

## Method

Every lane froze its criteria in a commit before any script ran, and every lane has a MUTATE twin that must fail. Footings are reported apart. No ΛCDM inputs were used. Scripts and outputs are in the repository under closure_2026_11_06/ and closure_2026_11_07/ at the commits the lane RESULT files name. The supplement archive deposited with this note holds every Lean source and build output, the closure tables, and each lane's criteria and result files, at the commit recorded as git_head in PAPER70_supplement_manifest.json, together with this note's source and audit script.